Diabetic Retinopathy Detection using Deep Learning

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ABSTRACT

Diabetic Retinopathy (DR) is an eye disease associated with chronic diabetes. DR is the leading cause of blindness among working aged adults around the world and estimated it may affect more than 93 million people. Progression to vision impairment can be slowed or controlled if DR is detected in time, however this can be difficult as the disease often shows few symptoms until it is too late to provide effective treatment.

Currently, detecting DR is a time-consuming and manual process, which requires an ophthalmologist or trained clinician to examine and evaluate digital color fundus photographs of the retina, to identify DR by the presence of lesions associated with the vascular abnormalities caused by the disease.

The automated method of DR screening will speed up the detection and decision-making process, which will help to control or manage DR progression. This paper presents an automated classification system, in which it analyzes fundus images with varying illumination and fields of view and generates a severity grade for diabetic retinopathy (DR) using machine learning models such as CNN, VGG-16 and VGG-19.

This system achieves 80% sensitivity, 82% accuracy, 82% specificity, and 0.904 AUC for classifying images into 5 categories ranging from 0 to 4, where 0 is no DR and 4 is proliferative DR.

CCS CONCEPTS

• **Computing methodologies → Neural networks**.

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KEYWORDS

Diabetic retinopathy, classification, image processing, deep learning, segmentation, severity grade

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1 INTRODUCTION

The American Diabetes Association conducted a study on diabetic retinopathy (DR) and observed that more than 4.4 million Americans of age 40 and older got DR during 2005-2008, with almost 0.7 million (4.4% of those with diabetes) having advanced DR that could lead to severe vision loss [6].

DR is one of the most common retinal diseases and it is a primary cause of blindness in humans [6]. It augments the blood pressure in small vessels and consequently influences the circulatory system of the retina and the light-sensitive tissue of the eye. Although the major risk factors for DR (e.g., hyperglycemia, hypertension, dyslipidemia) have been examined in many epidemiologic studies and clinical trials, there is considerable variation in the consistency, pattern, and strength of these risk factors.

Increasing life expectancy, indulgent lifestyles and other contributing factors mean the number of people with diabetes is projected to continue rising. Regular screening of diabetic patients for DR has been shown to be a cost-effective and important aspect of their care. The accuracy and timing of this care is of significant importance to both the cost and effectiveness of treatment. If detected early enough, effective treatment of DR is available, making this a vital process.

Classification of DR involves the weighting of numerous features and the location. This is highly time consuming for clinicians.

Computers are able to obtain much quicker classifications once trained, giving the ability to aid clinicians in real-time classification. The efficacy of automated grading for DR has been an active area of research in computer imaging. Convolutional Neural Networks (CNNs), a branch of deep learning [14], have an impressive record for applications in image analysis and interpretation, including medical imaging.

Flexibility and customizability in CNNs such as implementation of dropout, rectified linear units (ReLU) and the accompanying increase in computing power through graphical processor units (GPUs) helps in solving complex image recognition problems.

2 PROBLEM BACKGROUND

DR is a complication caused by diabetes due to damage of the blood vessels of the light-sensitive tissue at the back of the eye [6]. Too much sugar in the blood leads to blockage of the tiny blood vessels, cutting off its blood supply and the eye attempts to grow new blood vessels. Early DR is called non-proliferative diabetic retinopathy where new blood vessels are not growing yet, but the walls in the blood vessels weakens resulting in tiny bulges protruding from vessel walls and sometimes exudates due to leakages of fluid and blood into the retina. Larger vessels begin to dilate and results in irregular diameter. As the condition progresses, the blood vessel gets blocked and retina swells. This results in growth of abnormal new blood vessels in the retina. These new vessels can result in leakage and scar tissue formation in the eye, which can finally lead to retinal detachment and glaucoma. Figure 1 shows an image with various types of diabetic retinopathy conditions. Hence, detection of such exudates, scars and abnormal blood vessels are important diagnostic tasks in DR. Typical manual diagnosis takes 7 to 14 days as shown in Fig. 2 and needs specialized professionals input. The purpose of this study is to predict DR and do automated analysis by assigning a score of severity based on high-resolution retina images. This will save time in manual diagnosis as well as provide support where specialists are not available.

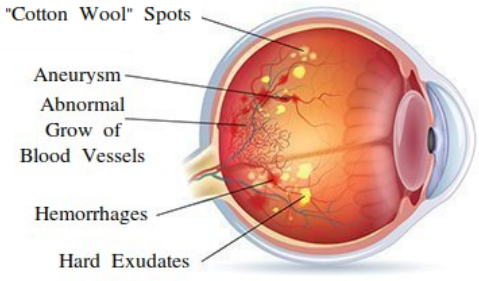


Figure 1: Diabetic retinopathy conditions

3 RELATED WORK

Previous effort on automatic DR detection was performed using machine learning methods like k-NN and Support Vector Machine (SVM) [9, 16, 17]. Acharya et al. [1] used features of blood vessel areas and SVM to classify the type of DR as microaneurysms, exudes, and hemorrhages and achieved an accuracy of 86% and specificity of 86%.

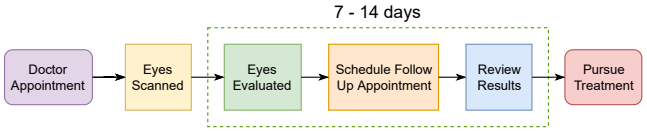


Figure 2: Manual diagnosis process

Recently, deep learning based architectures have become popular and showed good results. Kaggle also launched a competition [8] for DR detection in 2015 where the top ranked solutions used CNN-based models. In that competition, Kelvin [10] used a pre-trained Inception-V3 model as a baseline and retrained some modified final layers of the CNN with attention to get an accuracy of 60% on the test images. Meena [18] achieved an DR classification accuracy of 74% on the validation dataset using basic pre-processing and a CNN model. Graham [7] achieved the first rank with score of 84.9% and he used SparseConvNet CNN and a Random Forest (RF) model to combine predictions from images of the two eyes into a single prediction.

4 DATA SOURCE

We have used the EyePACS dataset [8] from Kaggle 2015 competition which contains coloured high-resolution retina images taken under a variety of imaging conditions. Images of left and right eyes are provided for every subject. Images are taken of different people for both left and right eyes. Images are taken using different types of cameras and are of different sizes and appearances. The data is extremely noisy and would require multiple preprocessing steps to get all images to a useable format for training a model. Figure 3 shows few sample of different appearances of the Retinal Fondue images. The purpose is to develop a robust algorithm that can function well in the presence of noise and image variation. Each image is rated on a scale of 0 to 4 levels for DR by clinician as below:

- `0` - No DR
- 1 - Mild
- 2 - Moderate
- 3 - Severe
- 4 - Proliferative DR

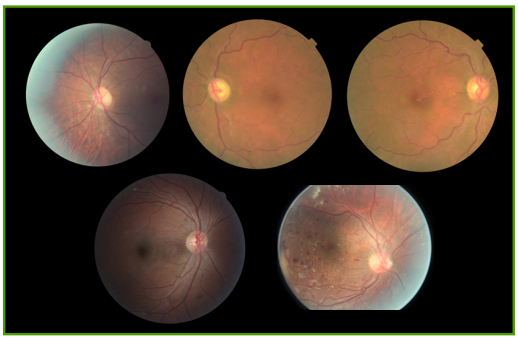


Figure 3: Image samples from the dataset

5 APPROACH

We will be adopting Meena's model [18] as the baseline and use various enhancing features to improve the performance. We plan to use following approaches to enhance the accuracy of the classification model:

(1) Data Transformation and Augmentation.
   (a) Crop the images to remove unnecessary noise.
   (b) Drop bad and irrelevant images as outliers.
   (c) Color normalization.
      - Convert images into black and white.
      - Enhance the images by highlighting spots in the images.
      - Subtracting the average color from images.
   (d) Mirror and rotate the images to create and augment the dataset.
(2) Add more layers: use deeper pre-trained networks.
(3) Reduce the labels to binary: better accuracy with high sensitivity due to the criticality of issue.
(4) Ensemble with traditional machine learning algorithms.

6 PROPOSED ARCHITECTURE

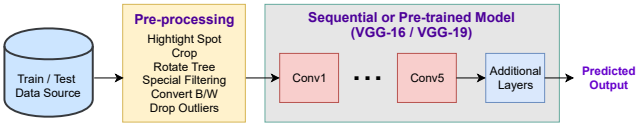


Figure 4: Our proposed approach

6.1 Data Exploration, Noisy Data Removal and Data Balancing

The training data comprised of 35,126 images of retinal fondue labelled as 5 different classes. During initial exploration of the image data, we found several noisy images that had to be dropped. The criteria for dropping bad images will be based on their colour spread in histogram.

Furthermore, the data showed that the image classes are highly imbalanced as shown in Fig. 5. There are 25,810 images of no DR types and 9,316 images are classified as having retinopathy of 4 different scales. To address the imbalance data, two steps were adopted:

- **Grouping of images as no DR and DR types**: We assigned all images with different scales of retinopathy as class 1, while those without retinopathy as class 0.
- **Data augmentation**: After preprocessing, all lower frequency class images will be augmented to get balanced data for images with different scales of retinopathy so the two classes gets balanced. We mirror and rotate the images to augment the data set.

6.2 Image Pre-processing

The images are rescaled to the same radius and cropped to remove background noise. Since there are a lot of exposure and lighting variation, the images needed to be standardized. First, a green filter is applied to get clarity in images, followed by the multi-channel

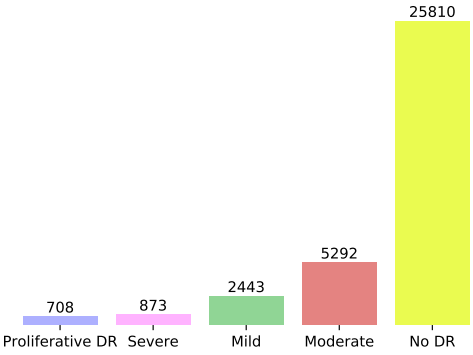


Figure 5: Class distribution of the images

images being converted to gray-scaled images and then normalized. Normalization will be done by dividing the converted grey-scale image intensities to 255. Images are cropped to remove noise in the black background around the retina fondue.

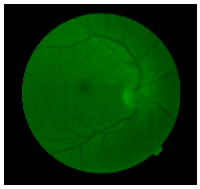


Figure 6: An image after pre-processing

6.3 Data Augmentation

Since the image classes are heavily imbalanced, we augment the training data to get balanced distribution among the DR severity classes. We mirror and rotate the images to create new augmented data set.

6.4 Modelling

Modelling was done using 2-D CNNs. Other than pre-processing, we tested and enhanced the model performance using pre-trained VGG-16 and VGG-19 CNN models. We used ensemble-based classifiers involving SVM and RF at the last stage of the classification layer.

As a result, we compared accuracy of VGG-16 and VGG-19, and as per our observations the accuracy is almost same for both models which ranges from 71 to 73%.

So, sequential modelling was attempted by adding layers manually and customizing the layers convolutional, dropout, dense, optimizers etc. Details of the layers are as in Fig. 7.

Convolutional layer.
This layer creates a convolution kernel that is convolved with the layer input over a single spatial (or temporal) dimension to produce a tensor of outputs. If bias is True, a bias vector is created and added to the outputs. Finally, if activation is not None, it is applied to the outputs as well.

When using this layer as the first layer in a model, provide an input shape argument (tuple of integers or None, does not include the

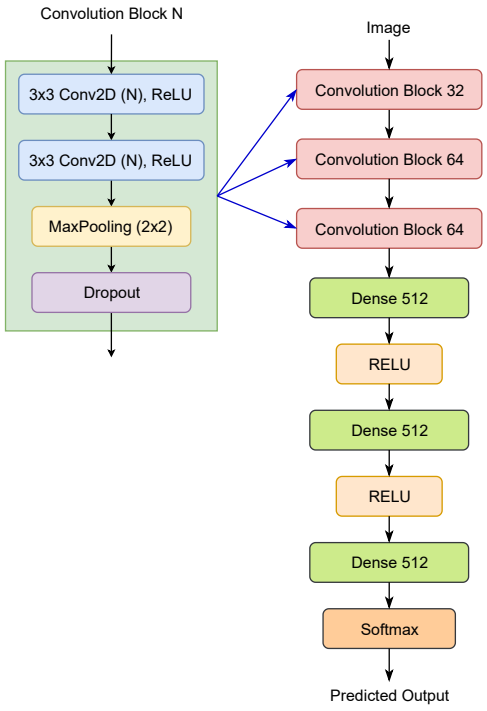


Figure 7: Details of the CNN layers

batch axis), e.g. input shape = (256,256,32) for time series sequences of 256 time steps with 256 features per step in data format="channels last", or (None, 256) for variable-length sequences with 256 features per step.

Pooling Layer.
This is one of the most significant layer which helps the network to avoid over-fitting by reducing the parameters and computation in the network. It works as a form of non-linear down sampling. Pooling partition the activation maps into sets of rectangles and collect the maximum value in the sub region. For instance, if $N \times N$ input layer, that will give output layer of $N/K \times N/K$ layer.

The main significance of this layer is to check whether a given feature is found anywhere in a region of the image. It then throws away the exact positional information.

Dropout layer.
Dropout works by probabilistically removing or "dropping out" inputs to a layer, which may be input variables in the data sample or activations from a previous layer. It has the effect of simulating a large number of networks with very different network structure and, in turn, making nodes in the network generally more robust to the inputs.

The crucial part of the deep convolutional neural network is handling the parameters generated from each stacked layer abundantly. It may cause over-fitting. For avoiding such scenarios, some neurons should be dropped in the layer which cascaded to the next layer. Dropouts are mainly near the fully-connected layer to avoid excessive generation of parameters. It is a widely used regularization techniques.

Flatten layer.
Flatten is used to reshape the tensor to such a shape which is equal to the number of elements present in the tensor.

Dense layer.
A dense layer is just a regular layer of neurons in a neural network. Each neuron receives input from all the neurons in the previous layer.

After we moved towards Keras sequential model, the accuracy was improved to 83%.

7 RESULTS AND DISCUSSION

We tested with VGG-16 and VGG-19 CNN models and observed that the accuracy always ranges from 71 to 73%. We then tested with Keras sequential CNN models by adding customized convolutional layer, optimiser, dense, merge etc. and obtained improved accuracies ranging from 80 to 83%. Due to the limitation of image label counts for 3 and 4 labels, the training model accuracy is impacted in labels 3 and 4. This can be improved if there is more samples with better distribution on each labels.

We can extend our work by including models built from using other deep learning architectures such as capsule neural networks [11, 15] and wide and deep learning [5, 12], and also by applying ensemble learning with different classifiers such as the support vector machines, kernel dictionary learning [2–4], enhanced k-NN [13], and gradient boosting, on the last feature map of each deep learning architecture. Ensemble learning often works well when there is a diversity of classifiers.

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